4.2 Position, Velocity and Acceleration — teacher key

Question: How do position, velocity and acceleration come from one another?

Self-paced lesson

No clock, no trimming, no skipping, no end mark, no stamp. Every tab is required in order; progress saves on the device so a student can stop and finish the next day.

Warm-up review (3 easy items)

- The derivative of $3t^2 - 12t$ is... **6t − 12.** Power rule: $3t^2$ gives $6t$, and $-12t$ gives -12.
- The derivative of $\sin t$ is... **cos t.** The derivative of sine is cosine.
- A line with a negative slope is... **going down from left to right.** Negative slope means the line falls as you move right.

Explore stations

Four hands-on stations (X1, X2, X3, X4). Station answers are in the teacher view of the lesson page (add `#key` to its address).

Explore: what did you see

- In Station 1, at the moment the particle was at rest, the velocity reading was... **0.** At rest means not moving, so the velocity is 0.
- In Station 2, the particle slowed down whenever the green arrow (velocity) and the orange arrow (acceleration) pointed... **opposite ways.** Opposite signs for v and a means the particle slows down.

Mid-lesson check (Huddle tab)

- $v(t) = -5$ and $a(t) = -2$ at the same moment. The particle is... **speeding up.** Both are negative, the same sign, so it is speeding up (in the negative direction).
- Acceleration $a(t)$ is the derivative of... **the velocity.** Acceleration is the derivative of velocity, which is the second derivative of position.

Targets

- I can find velocity and acceleration from a position function.
- I can say which way a particle moves from the sign of v.
- I can tell if a particle speeds up or slows down.

Predictions

- A particle has position $s(t)$ meters at t seconds. Its velocity is... **the derivative $s'(t)$.** Velocity is how fast the position changes: the derivative of position.
- A particle moves on a horizontal line, with right as the positive direction. Its velocity is -4 meters per second. It is moving... **left.** Negative velocity means moving in the negative direction: left.

Huddle script (answers in green)

Definitions

- **Velocity and acceleration:** v(t) = s?(t). a(t) = **v?(t)**, the derivative of velocity, so a(t) = s?(t).
- **Direction:** If v > 0 the particle moves **right**. If v < 0 it moves **left**. If v = 0 it is **at rest**.
- **Speed:** Speed is the **absolute value** of velocity: speed = |v|.
- **Speeding up or slowing down:** v and a with the **same** sign: speeding up. With **opposite** signs: slowing down.

Worked example

A particle has position $s(t) = t^3 - 6t^2 + 9t$. Find v and a, then describe t = 2 and t = 4.

- Velocity: $v(t) = 3t^2 - 12t + \mathbf{9}$
- Acceleration: $a(t) = \mathbf{6t - 12}$
- At t = 2: v = –3 and a = **0**, so it moves left at steady speed for an instant.
- At t = 4: v = 9 and a = 12, both **positive**, so it is speeding up.

Label the picture

1. The dot: the particle at position **s**.
2. The green arrow: the **velocity**.
3. The orange arrow: the **acceleration**.

Every level, with answers

Find the velocity

Level 1. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $9t^2 + 8t + 4$ $v(t) = s'(t) = 9t^2 + 8t + 4$.

Level 2. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $8\cos(4t)$ $v(t) = s'(t) = 8\cos(4t)$.

Level 3. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $4t\left(t^2 + 1\right)$ $v(t) = s'(t) = 4t(t^2 + 1)$.

Level 4. A particle moves along a line with position s in meters at time t seconds. Find its velocity.
Answer: $(1 + 3t)\, e^{3t}$ $v(t) = s'(t) = (1 + 3t)e^{3t}$.

Find the acceleration

Level 1. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $12t$ $v(t) = 6t^2 + 7$. Differentiating v gives the acceleration, the answer above.

Level 2. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $(-24)\cos(2t)$ $v(t) = -12\sin(2t)$. Differentiating v gives the acceleration, the answer above.

Level 3. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $\left(2+4t^2\right)e^{t^2}$ $v(t) = 2te^{t^2}$. Differentiating v gives the acceleration, the answer above.

Level 4. A particle moves along a line with position s in meters at time t seconds. Find its acceleration.
Answer: $2\cos(t) - t\sin(t)$ $v(t) = \sin t + t\cos t$. Differentiating v gives the acceleration, the answer above.

Which way is it moving?

Level 1. A particle moves on a horizontal line, with right as the positive direction. At $t = 3$ seconds its velocity is -2 meters per second. Which way is it moving?
(A) to the left (B) to the right (C) cannot tell (D) at rest for an instant
Answer: A The velocity is -2, so the particle is to the left.

Level 2. A particle moves on a horizontal line, with right as the positive direction. Its velocity is $v(t) = 3t^2 + 4t - 1$ meters per second. At $t = 2$ which way is it moving?
(A) to the left (B) at rest for an instant (C) cannot tell **(D) to the right**
Answer: D $v(2) = 19$, so the particle is to the right.

Level 3. A particle moves on a horizontal line, with right as the positive direction. Its position is $s(t) = t^3 - 4t^2 - 4t + 5$ meters. At $t = 4$ which way is it moving?
(A) at rest for an instant (B) cannot tell (C) to the left **(D) to the right**
Answer: D $v(t) = 3t^2 - 8t - 4$ and $v(4) = 12$, so the particle is to the right.

Level 4. A particle moves on a horizontal line, with right as the positive direction. Its position is $s(t) = 3t^3 - 8t^2 + 11t + 5$ meters. At $t = 4$ which way is it moving?
(A) cannot tell (B) to the left (C) at rest for an instant **(D) to the right**
Answer: D $v(t) = 9t^2 - 16t + 11$ and $v(4) = 91$, so the particle is to the right.

Speeding up or slowing down?

Level 1. At one moment a particle has velocity $v = -3$ and acceleration $a = -4$ (meters per second, and meters per second squared). Is it speeding up or slowing down?
(A) speeding up (B) slowing down (C) neither: its speed is not changing at that moment
Answer: A $v = -3$ and $a = -4$ have the same sign, so it is speeding up.

Level 2. A particle has velocity $v(t) = 3t^2 + 2t - 1$ meters per second. At $t = 0$ is it speeding up or slowing down?
(A) neither: its speed is not changing at that moment **(B) slowing down** (C) speeding up
Answer: B $v(0) = -1$ and $a(0) = 2$. Opposite signs: slowing down.

Level 3. A particle has position $s(t) = t^3 - 4t^2 - 4t + 5$ meters. At $t = 5$ is it speeding up or slowing down?
(A) slowing down **(B) speeding up** (C) neither: its speed is not changing at that moment
Answer: B $v(5) = 31$ and $a(5) = 22$. Same sign: speeding up.

Level 4. A particle has position $s(t) = 3t^3 - 8t^2 + 11t + 5$ meters. At $t = 4$ is it speeding up or slowing down?
(A) slowing down **(B) speeding up** (C) neither: its speed is not changing at that moment
Answer: B $v(4) = 91$ and $a(4) = 56$. Same sign: speeding up.

Common misconceptions

Students say	Say instead
Negative acceleration means the particle is slowing down.	It depends on v. Negative a slows a particle moving right, but speeds one moving left.
Velocity and speed are the same.	Speed is $\|v\|$. A particle moving left at 3 has velocity –3 and speed 3.
If $v = 0$, the particle stops for good.	It may only be at rest for an instant, then turn around.
If $v = 0$, then $a = 0$.	Not so. A ball at the top of its path has $v = 0$ but a is not 0.
$a(t) = s?(t)$.	$v = s?$ and $a = v? = s?$. Acceleration takes two derivatives of position.